

Liquid- Liquid Extraction

Pre-Lab Plan
Unit Operations Lab #3
02/20/20

Group #2, Thursday, AM section

Team member Names

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Leena Daniel

Jay Rawal

Esteban Vazquez- Elizondo

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1. Experimental Objective

The objective of this lab is to operate a three-stage mixer-settler system to extract acetic acid from white oil by using water as the solvent and to investigate the effect of the speed of mixing on the efficiency of extraction on the unit.

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Mention that you'll use McCabe-Thiele analysis

Commented [AZ3]: This is the objective (along with the McCabe Thiele analysis). Operating the unit is the means to the objective.

2. Experimental

2.1 Apparatus

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Missing pump switch

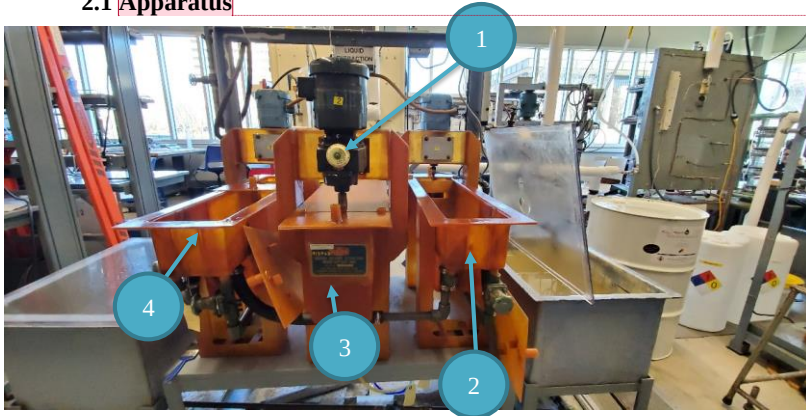


Figure 1: Overview of the Liquid Liquid Extraction Unit



Figure 2: Product Tank

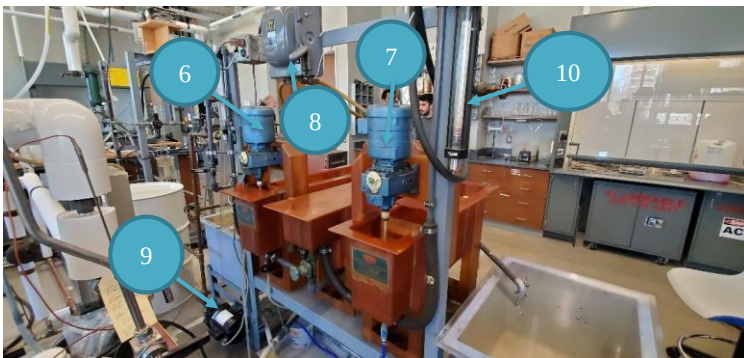


Figure 4: Backview of the Liquid Liquid Extraction Unit



Figure 3: Mixer tank

Number	Equipment	Description
1	Stage 2 Agitator	Mixes the solution in the 2 nd stage tank
2	Stage 1 Tank	1 st stage tank that extract the white oil from the acetic acid
3	Stage 2 Tank	2 nd stage tank that extracts the white oil from the acetic acid
4	Stage 3 Tank	3 rd stage tank that extracts the white oil from the acetic acid
5	Raffinate Reservoir	Contains the acetic acid, which is the raffinate from each of the stages
6	Stage 1 Agitator	Mixes the solution in the 1 st stage tank
7	Stage 3 Agitator	Mixes the solution in the 3 rd stage tank
8	Agitator Power Switch	Turns on all of the agitator
9	Pump	Adds the solutions into the different tanks
10	Water Rotameter	Used to set the water flowrate
11	Feed Solution Reservoir	The Feed is added in and mixed through this reservoir
12	Oil Rotameter	Used to set the oil flowrate

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2.2 Materials and Supplies

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Materials/Supplies	Brief Description
Graduated Cylinder	To measure the flowrates of the extract and the raffinate phases
Phenolphthalein Indicator	To be used to indicate the amount of NaOH required in titration
Standard NaOH solution of 0.05N	To be used to titrate the sample to determine the composition of the mixture
50mL Burette and Burette Clamp	Used to hold the NaOH solution used for titration
125mL Erlenmeyer flasks	Used to collect the samples needed for titration
Parafilm	To cover the samples
Pipet and Bulb	To extract the samples from the tank into the flask
Water	Used as a solvent
White Oil	It has different density than the water and is used to dilute the acetic acid
Acetic Acid	Used in the feed mixture along with the white oil
Stir Bar and Stir Plate	Used to mix the solution being titrated

Commented [T7]: Missing table caption and tachometer to measure the agitator speed

2.3 Experimental Plan

1. In this experiment the dependent variables are the acetic acid in the extract and in the raffinate while the agitator speeds are the independent variables.
2. Make sure that all the reflux valves are fully closed in each step
3. If the white oil is in the product tank, send it back to the feed tank to level it to about 25 gallons
4. Perform a titration to check whether there are trace amount of acetic acid in the mixture
5. Prepare a mixture of 25 gallons of white oil and acetic acid which will have 0.5 wt% of Acetic acid
6. Mix thoroughly in the feed tank and operator using the centrifugal pump
7. Take the sample to determine whether the composition is accurate by titrating it using a standard NaOH solution and phenolphthalein indicator
8. Establish the water circuit in the mixer-settler unit to have steady levels and adjust the speed of the agitator to a value between 300 RPM and 1000 RPM
9. Feed in the white oil and adjust the water flowrate to be 0.67 GPM and the oil flowrate to be 0.34 GPM. The ratio between the solvent and the feed flowrate should be at least 2
10. Steady levels must be reached at the two phases for each stage and shall remain constant during the experiment
11. Choose a white oil flowrate that can conduct 3 runs using the 25 gallons and allow the system to reach steady state
12. Take samples of the white oil and the water phases from the 3 stages and titrate the samples to determine the composition
13. Repeat 7-11 using two other agitator speeds while the flowrates are kept constant

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Missing dependent and independent variables

Commented [T9]: How much acetic acid? In ml?

Commented [T10]: Which speeds are you planning to use?

Project Safety Evaluation

Commented [T11]: 9.5/10

Project Title: Liquid-Liquid Extraction

Date: 02/20/20

	<i>Yes/No</i>	
Potential electrical hazards?	Yes	If yes, identify conditions: The agitators and pumps are electric. Make sure to handle fluids with care.
Potential mechanical hazards?	Yes	If yes, identify conditions: Make sure the agitator speeds do not go under 300 RPM and above 1000 RPM
Potential pressure hazards?	No	If yes, identify conditions:
Potential temperature hazards?	No	If yes, identify conditions:
Hazardous/toxic chemicals involved?	Yes	Acetic acid can cause skin irritation and is also flammable and corrosive Phenolphthalein can cause skin/eye irritation and is also flammable Sodium hydroxide can cause skin irritation
Gloves required?	Yes	If yes, specify type: Nitrile Gloves
MSDS reviewed by all team members?	Yes	8042-47-5 White Oil 64-19-7 Acetic acid 1310-73-2 Sodium hydroxide 77-09-8 Phenolphthalein 7732-18-5 Water
<i>Process Parameters</i>	<i>Max Expected</i>	<i>List location and possible hazards and precautions</i>
Temperature (°C)	25°C	Room Temperature
Pressure (psia)	14.2	Room pressure
<i>Safety Controls</i>	<i>yes/no</i>	<i>If yes, Provide description</i>
Agitator Power Switch	Yes	Stops all the agitators

Commented [T12]: And pump switch for shutting off the pump

I have read relevant background material for the Unit Operations Laboratory entitled: “Liquid-Liquid Extraction” and understand the hazards associated with conducting this experiment. I have planned out my experimental work in accordance to standards and acceptable safety practices and will conduct all of my experimental work in a careful and safe manner. I will also be aware of my surroundings, my group members, and other lab students, and will look out for their safety as well.

Diana Awimrin	02/20/20
signature of team member	date
Leena Daniel	02/20/20
signature of team member	date
Jay Rawal	02/20/20
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